

AFYONKARAHISAR, Turkey

WMO: 171890

Lat: 38.733N

Lon: 30.600E

Elev: 1001

StdP: 89.86

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-8.8	-14.2	1.2	-10.3	-11.6	1.6	-7.4	11.2	6.5	9.2	5.1	1.3	310	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	33.5	17.0	31.9	16.8	30.3	16.5	18.9	29.0	18.2	28.1	17.5	27.3	3.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	12.6	22.3	14.8	11.9	21.7	13.9	11.2	21.2	58.0	28.8	55.4	28.0	53.3	27.3	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.5	6.3	-16.2	35.7	5.4	1.8	-20.0	37.0	-23.2	38.1	-26.2	39.1	-30.1	40.4		
(5)			-16.5	20.2	5.2	1.2	-20.3	21.1	-23.3	21.8	-26.2	22.4	-30.0	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	0.7	2.6	6.1	10.4	15.0	18.9	22.4	22.4	18.1	12.7	7.0	2.5	(6)
(7)		DBStd	8.46	5.18	4.97	4.27	4.00	3.31	2.87	2.43	2.52	3.12	3.76	4.23	4.68	(7)
(8)		HDD10.0	1033	288	211	132	41	3	0	0	0	0	18	106	235	(8)
(9)		HDD18.3	2794	546	442	379	238	111	27	1	2	41	177	339	491	(9)
(10)		CDD10.0	1624	1	2	11	54	158	267	386	384	243	102	16	2	(10)
(11)		CDD18.3	344	0	0	0	1	7	44	128	128	34	3	0	0	(11)
(12)		CDH23.3	4372	0	0	1	21	160	617	1534	1474	521	46	0	0	(12)
(13)		CDH26.7	1629	0	0	0	1	23	183	646	616	157	4	0	0	(13)
(14)	Wind	WSAvg	2.4	2.2	2.4	2.6	2.5	2.4	2.6	2.8	2.6	2.3	2.0	1.9	2.1	(14)
(15)	Precipitation	PrecAvg	467	47	43	47	52	51	35	19	17	20	40	38	54	(15)
(16)		PrecMax	718	126	107	101	89	114	159	85	71	84	90	90	169	(16)
(17)		PrecMin	312	6	7	12	25	2	2	0	1	0	1	5	5	(17)
(18)		PrecStd	78	26	25	23	19	27	33	19	18	22	23	23	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	17.2	20.7	25.6	29.1	32.6	36.0	35.8	32.5	27.0	20.3	15.9	(19)
(20)			MCWB	7.2	8.8	9.7	12.9	15.9	16.5	17.1	17.8	15.6	14.3	10.9	8.2	(20)
(21)		2%	DB	11.0	13.8	18.0	23.1	26.9	30.7	33.7	33.7	30.2	24.5	18.0	12.7	(21)
(22)			MCWB	6.5	7.3	8.8	11.9	14.6	16.5	17.3	17.1	15.4	13.4	10.6	7.7	(22)
(23)		5%	DB	9.1	11.7	15.9	20.6	25.1	28.8	32.0	31.9	28.4	22.3	15.9	10.7	(23)
(24)			MCWB	5.6	6.4	7.9	11.1	14.1	16.0	17.0	16.8	14.9	12.6	9.6	7.0	(24)
(25)		10%	DB	7.5	9.7	13.7	18.2	23.2	27.0	30.3	30.1	26.5	20.1	14.0	9.0	(25)
(26)			MCWB	4.7	5.7	7.2	10.1	13.4	15.4	16.6	16.6	14.6	12.0	8.7	6.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	9.3	10.7	13.8	17.5	18.9	20.1	19.8	18.0	15.8	12.5	9.5	(27)
(28)			MCDWB	12.0	15.6	18.5	23.1	26.3	28.4	30.7	29.8	28.5	24.3	17.8	13.7	(28)
(29)		2%	WB	6.7	7.8	9.4	12.5	15.7	17.7	19.0	18.9	16.8	14.4	11.1	8.2	(29)
(30)			MCDWB	10.4	12.9	16.4	21.2	24.3	27.1	29.6	28.9	26.3	21.5	16.5	11.9	(30)
(31)		5%	WB	5.7	6.7	8.5	11.5	14.8	16.9	18.1	18.2	16.1	13.5	10.1	7.2	(31)
(32)			MCDWB	8.7	11.1	14.8	19.3	23.1	26.0	28.4	28.0	25.1	20.1	14.9	10.5	(32)
(33)		10%	WB	4.8	5.7	7.5	10.6	13.9	16.2	17.5	17.6	15.3	12.7	9.1	6.2	(33)
(34)			MCDWB	7.3	9.3	13.0	17.4	21.7	24.9	27.5	27.1	24.0	18.9	13.2	8.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	8.8	10.3	11.5	13.1	13.9	14.9	14.8	14.6	12.2	10.9	7.6	(35)
(36)			MCDBR	9.5	12.2	14.5	15.6	16.5	17.2	17.2	17.4	18.1	16.0	13.9	10.5	(36)
(37)			MCWBR	6.0	7.0	7.3	7.0	7.0	6.1	5.4	5.2	6.9	7.3	7.6	6.6	(37)
(38)		5% WB	MCDWB	8.7	11.1	12.9	13.8	15.0	14.8	15.1	14.7	15.0	13.2	11.9	9.3	(38)
(39)			MCWBR	5.7	6.6	6.9	6.7	6.9	6.1	5.4	5.0	6.1	6.6	6.8	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.293	0.320	0.364	0.394	0.380	0.359	0.355	0.358	0.338	0.335	0.307	0.294	(40)	
(41)		taud	2.424	2.370	2.267	2.210	2.309	2.388	2.393	2.366	2.422	2.429	2.474	2.477	(41)	
(42)		Ebn at Noon	887	903	891	883	899	916	916	907	908	875	863	860	(42)	
(43)		Edh at Noon	83	100	124	139	129	119	118	118	105	94	79	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.04	2.93	4.10	5.26	6.25	7.20	7.61	6.75	5.37	3.74	2.61	1.84	(44)	
(45)		RadStd	0.23	0.40	0.32	0.42	0.47	0.54	0.28	0.23	0.36	0.24	0.27	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	-0.52	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page